

CURRICULUM VITAE

Personal data

Axel Gabriel Rodríguez Zárte

axel.rodriguez@cimat.mx

Website: gabiroduz.github.io (work in progress!)

Education

- Bachelor degree in Mathematics from the University of Guanajuato, in collaboration with the Centro de Investigación en Matemáticas (CIMAT, A.C.). In administrative process.
- High School at Tecnológico de Monterrey IB Programme, funded by an academic excellence scholarship.

Languages

- Spanish, native.
- English:
 - TOEFL score of 110 (5.5 of 6 in the 2026 scale, with 6 of 6 in reading, 5.5 of 6 in listening, 5.5 of 6 in speaking and 5.5 of 6 in writing).
 - Advanced, fluid, conversational and technical.

Professional skills

- Python, C, C++.
- Github.
- Experience in Web Development (PHP Laravel, Javascript, CSS/HTML)
- Fast and eager learner. I can get up to speed very quickly.
- Problem solving mentality and analytical thinking, as a successful Mathematics student.
- Excel.
- Comfortable with Linux-based Operating Systems.
- High proficiency with technology in general. (i.e. Understanding computer networks, solving basic technical issues)
- Familiarity with Cloud services (AWS, Azure).

Academia

Teaching experience

- Sensorium Artes Contemporáneas: I have taught Calculus, and Statistics under the COBAEJ study plan, and Pensamiento Matemático I, II, III for the BiS study plan with autonomously planned lessons and self-written manuals.

Also taught the Science Workshop II and the Ecosystems class. Since September 2025.

Academic interests

- Geometrical methods of mathematical physics.
- Applications (Material informatics, Condensed Matter Physics, Geometric Deep Learning)

Academic activity

- Teaching Assistant, University of Guanajuato, Department of Mathematics.
 - Class: Introduction to Computer Science.
 - Professors: Dr. Alonso Ramírez Manzanares, Dr. Jean-Bernard Hayet. 2022-08 / 2022-12.
- Teaching Assistant, University of Guanajuato, Department of Mathematics.
 - Class: Introduction to Computer Science.
 - Professors: Dr. Alonso Ramírez Manzanares, Dr. Jean-Bernard Hayet. 2023-08 / 2023-12.
- Teaching Assistant, University of Guanajuato, Department of Mathematics.
 - Class: Calculus II.
 - Professor: Dr. Ehyter M. Martín González. 2024-01 / 2024-06.
- Teaching Assistant, University of Guanajuato, Department of Mathematics.
 - Class: Calculus I.
 - Professor: Dr. Ehyter M. Martín González. 2024-06 / 2024-12.
- Summer stay in CIMAT, A.C. (Epidemiologic modeling)
 - Professors: Dr. Uvencio José Giménez Mujica, Dr. Oziel Gómez Martínez. 2024-06 / 2024-08
- Teaching Assistant, University of Guanajuato, Department of Mathematics.
 - Class: Calculus IV
 - Professor: Dr. Luis Hernández Lamóneda. 2025-01 / 2025-06
 - Bibliography: Spivak, Calculus on Manifolds. Marsden, Vector Calculus. Arnold, Mathematical Methods of Classical Mechanics.

Grants and scholarships.

- Academic excellence stipend by Centro de Investigación en Matemáticas (CIMAT, A.C.). 2021-06 / 2025-06.
- Teaching assistant stipend by Centro de Investigación en Matemáticas

(CIMAT, A.C.). 2022-08 / 2022-12. 2023-08 / 2023-12. 2024-06 / 2025-06

- Lodging and food by UNAM Department of Mathematics, Juriquilla, for the Summer School 2024.

Other activity

- Summer School 2024, UNAM Juriquilla. Biomathematics 2024-06
- Summer School 2024, CIMAT. Low-dimensional topology and other selected topics.
- Summer School 2024, CINVESTAV. Selected Analysis topics.
- MSc level class in Fourier Analysis (2025/06 - 2025/12) at CIMAT.